Gaia DR3 5870569352746778624: A $\sim 12.5\,\mathrm{M}_\odot$ Stellar-Mass Black Hole Candidate in the Galactic Disk from Combined Astrometric and Spectroscopic Orbital Solution

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ABSTRACT

We present a multi-wavelength analysis of Gaia DR3 5870569352746778624, a binary system in the DR3 non-single-star (NSS) catalogue with a combined astrometric and spectroscopic orbital solution (`AstroSpectroSB1`) — the strongest solution type available, providing independent confirmation from both photo-centre motion and radial-velocity variability. Under the Gaia NSS pipeline assumptions — including an adopted primary mass $M_1 = 1.04\,\mathrm{M}_\odot$ — the derived companion mass is $M_2 = 12.55\,\mathrm{M}_\odot$, placing it firmly in the stellar-mass black hole regime. Our Monte Carlo posterior ($5{\times}10^5$ draws), propagating uncertainties in M_1, P, and ϖ, yields $M_2 = 12.5^{+7.3}_{-4.0}\,\mathrm{M}_\odot$ (median and 68 per cent credible interval $[8.5, 19.8]\,\mathrm{M}_\odot$), with $P(M_2 > 5\,\mathrm{M}_\odot) = 99.6$ per cent and $P(M_2 > 10\,\mathrm{M}_\odot) = 71.4$ per cent. Inflating the DR3 parallax uncertainty by a factor of 1.5× (as recommended for orbital solutions near $G \approx 12$) still yields $P(M_2 > 5\,\mathrm{M}_\odot) = 97$ per cent. The system comprises a K2–K3 III giant primary ($G = 12.28$, $G_{\mathrm{BP}} - G_{\mathrm{RP}} = 1.49$; Gaia GSP-Phot $T_{\mathrm{eff}} = 4832\,\mathrm{K}$, $\log g = 3.26$) and an unseen companion orbiting with $P = 1352.3\,\mathrm{d}$ ($\sim 3.7\,\mathrm{yr}$), $e = 0.532$, and $a \approx 5.7\,\mathrm{AU}$ at $d \approx 1490\,\mathrm{pc}$. The 39.9σ NSS detection significance — 8× the acceptance threshold — combined with the dual astrometric + spectroscopic signal makes this one of the most statistically robust BH candidates in the Gaia sample. A main-sequence star of $12.55\,\mathrm{M}_\odot$ would be a late-B type star with $L \sim 17{,}000\,\mathrm{L}_\odot$ and $T_{\mathrm{eff}} \sim 24{,}000\,\mathrm{K}$, outshining the $\sim 47\,\mathrm{L}_\odot$ K-giant primary by a factor of ~ 360 bolometrically and $\sim 42\times$ in the G band — firmly excluded by the observed single cool-star SED. All seven non-BH alternative scenarios are excluded: five on mass or luminosity grounds (MS companion, WD, NS, hierarchical triple, stripped He star) and two by the combined astrometric + spectroscopic confirmation (astrometric artefact, chance alignment). The source lies at Galactic latitude $b = 2.8°$ with $|z| \approx 72\,\mathrm{pc}$, firmly in the Galactic thin disk. The measured radial velocity ($v_{\mathrm{sys}} = -5.04 \pm 2.08\,\mathrm{km\,s^{-1}}$) is consistent with normal disk kinematics. With $M_2 \sim 12.5\,\mathrm{M}_\odot$, this candidate is comparable to Gaia BH1 ($9.6\,\mathrm{M}_\odot$) and Gaia BH2 ($8.9\,\mathrm{M}_\odot$), and among the most massive stellar-mass BH candidates from Gaia with AstroSpectroSB1 confirmation. We note that Tanikawa et al. (2023) independently identified this source as a BH candidate; our analysis provides the first detailed multi-wavelength characterisation and uncertainty quantification.

Key words: binaries: astrometric – binaries: spectroscopic – stars: black holes – astrometry – stars: individual: Gaia DR3 5870569352746778624

1 INTRODUCTION

Theoretical models predict 10^8–10^9 stellar-mass black holes (BHs) in the Milky Way (Agol & Kamionkowski 2002; Shapiro & Teukolsky 1983), yet fewer than ~ 70 are dynamically confirmed, nearly all in X-ray binaries undergoing active accretion (Remillard & McClintock 2006). The vast majority are expected to reside in non-interacting (dormant) systems, producing no electromagnetic signature beyond gravitational influence on a luminous companion.

Gaia Data Release 3 (DR3; Gaia Collaboration et al. 2023a) has transformed the search for dormant BHs through its non-single-star (NSS) catalogue (Gaia Collaboration et al. 2023b), which provides orbital solutions for $\sim 180\,000$ binaries. The landmark discoveries of Gaia BH1 ($M_2 \approx 9.6\,\mathrm{M}_\odot$; El-Badry et al. 2023a), Gaia BH2 ($M_2 \approx 8.9\,\mathrm{M}_\odot$; El-Badry et al. 2023b), and Gaia BH3 ($M_2 \approx 33\,\mathrm{M}_\odot$; Gaia Collaboration et al. 2024) have established the efficacy of astrometric BH detection. Notably, Gaia BH1 and BH2 were discovered using `AstroSpectroSB1` solutions — the most powerful solution type in the NSS catalogue, combining independent astrometric and spectroscopic constraints on the orbit.

The `AstroSpectroSB1` solution type is critically important because it provides *two independent lines of evidence* for the

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orbital signal. Pure astrometric (`Orbital`) solutions can be affected by systematic biases in the photo-centre astrometry, but when the same Keplerian orbit is independently detected in radial-velocity variations, the probability of a spurious artefact drops dramatically. El-Badry et al. (2024) showed that $\sim$20 per cent of pure `Orbital` solutions fail independent RV checks, but this failure rate is much lower for `AstroSpectroSB1` solutions where the RV signal is already built into the fit.

In a companion paper (Ayala-Baez 2026, in prep.; GRAVITAS OmniScan v13 catalogue) we identify 614 compact-object candidates from the Gaia DR3 NSS catalogue, 163 of which are GOLD-tier BH candidates. In this paper we focus on Gaia DR3 5870569352746778624, a GOLD-tier candidate with a derived companion mass of $M_2 = 12.55\,\mathrm{M}_\odot$ from an `AstroSpectroSB1` solution. Tanikawa et al. (2023) independently identified this source as a BH candidate in their astrometric binary search. Our analysis provides a deeper characterisation: Monte Carlo uncertainty propagation (including parallax), multi-wavelength SED checks, companion-light exclusion, alternative-scenario assessment, and archival upper limits. At $M_2 \sim 12.5\,\mathrm{M}_\odot$, this is comparable to the confirmed Gaia BH1 ($9.6\,\mathrm{M}_\odot$) and BH2 ($8.9\,\mathrm{M}_\odot$), and exceeds both by $\sim$30–40 per cent.

The system is a wide binary ($P = 1352.3\,\mathrm{d} \approx 3.7\,\mathrm{yr}$, $e = 0.532$) comprising a K2–K3 III giant primary ($M_1 \approx 1.04\,\mathrm{M}_\odot$) and a dark companion at $d \approx 1490\,\mathrm{pc}$ in the Galactic thin disk ($b = 2.8°$, $|z| \approx 72\,\mathrm{pc}$). The M_2/M_1 ratio of $\sim$12:1 means that M_2 has a moderate dependence on the assumed M_1, but the mass remains firmly above the BH threshold ($> 5\,\mathrm{M}_\odot$) for all plausible primary masses.

Section 2 describes the observational data. Section 3 presents the analysis methodology. Section 4 reports results. Section 5 tests non-BH explanations. Section 6 discusses implications and Section 7 summarises our findings.

2 DATA

2.1 Gaia DR3 astrometry and NSS orbital solution

Gaia DR3 5870569352746778624 appears in both the main astrometric catalogue and the NSS orbital supplement with a combined astrometric and spectroscopic solution (model type: `AstroSpectroSB1`). This solution type simultaneously fits the photo-centre orbit from epoch astrometry *and* the radial-velocity curve from the Gaia Radial Velocity Spectrometer (RVS), providing two independent constraints on the binary orbit.

Key parameters include parallax $\varpi = 0.672 \pm 0.104\,\mathrm{mas}$ ($d \approx 1488\,\mathrm{pc}$; 15.5 per cent relative error), photometry $G = 12.277$, $G_{\mathrm{BP}} = 12.965$, $G_{\mathrm{RP}} = 11.472$ ($G_{\mathrm{BP}} - G_{\mathrm{RP}} = 1.493$), and RUWE = 9.22. The astrometric excess noise significance is 1762σ.

The extremely elevated RUWE and very large excess noise significance are consistent with genuine binary photo-centre motion. RUWE = 9.22 is extreme even by binary standards, reflecting the combination of a massive dark companion and the relatively bright ($G = 12.3$) primary. The excess noise significance of 1762σ is among the highest in the NSS catalogue.

The NSS orbital solution yields: period $P = 1352.29 \pm 45.47\,\mathrm{d}$ (3.4 per cent precision), eccentricity $e = 0.532 \pm 0.015$, and solution significance 39.85σ — 8× the acceptance threshold and among the highest in the BH candidate sample. The goodness-of-fit is 3.07, acceptably close to zero. The derived companion mass $M_2 = 12.55\,\mathrm{M}_\odot$ from the Gaia `binary_masses` table assumes the pipeline's adopted primary mass of $M_1 = 1.04\,\mathrm{M}_\odot$.

The Gaia GSP-Phot pipeline produced atmospheric parameters for this source: $T_{\mathrm{eff}} = 4832\,K$ *and* $\log g = 3.255$ *dex.* This places the primary as a K2–K3 giant, well-constrained by the spectrophotometric fit.

A systemic radial velocity of $v_{\mathrm{sys}} = -5.04 \pm 2.08\,\mathrm{km\,s^{-1}}$ is available from the Gaia RVS.

2.2 Multi-wavelength photometry

Table 1 lists the photometry used in this work. At $G = 12.3$ the source is well within the optimal range of Gaia's photometric and spectroscopic pipelines. The Gaia photometry provides $G = 12.277$, $G_{\mathrm{BP}} = 12.965$, $G_{\mathrm{RP}} = 11.472$ ($G_{\mathrm{BP}} - G_{\mathrm{RP}} = 1.493$).

2.3 X-ray archival data

We find no ROSAT (2RXS; Boller et al. 2016) or XMM-Newton (4XMM-DR13; Webb et al. 2020) counterpart within 30″. The eROSITA eRASS1 catalogue (Merloni et al. 2024) covers the western Galactic hemisphere ($180° < l < 360°$); at $l = 310.4°$ the source falls within the eROSITA footprint, yet no counterpart is catalogued. Adopting a typical 0.5–2 keV sensitivity limit of $\sim 5 \times 10^{-14}\,\mathrm{erg\,cm^{-2}\,s^{-1}}$ yields an X-ray luminosity upper limit $L_{\mathrm{X}} < 5.3 \times 10^{30}\,\mathrm{erg\,s^{-1}}$ at $d = 1490\,\mathrm{pc}$, consistent with a quiescent non-accreting BH (cf. Gaia BH1/BH2; El-Badry et al. 2023a,b).

In the radio, the source lies within the RACS-low and SUMSS survey footprints (Dec $< -30°$) but is not detected in either catalogue. The VLASS footprint does not cover Dec $\approx -59°$. At $b = 2.8°$, moderate Galactic absorption (N_{H}) is expected, which may suppress soft X-ray emission. However, the absence of X-ray and radio emission is consistent with a quiescent, non-accreting system. At periastron, the separation $a(1-e) \approx 2.7\,\mathrm{AU}$ is much larger than the Roche lobe of the K-giant primary, so active mass transfer is not expected.

2.4 SIMBAD and literature cross-identification

Querying SIMBAD (Wenger et al. 2000) at the source position returns no prior compact-object classification for this source. Tanikawa et al. (2023) identified it as a BH candidate in their systematic astrometric-binary search of Gaia DR3, but no detailed characterisation has been published. Our work provides the first comprehensive multi-wavelength analysis and uncertainty quantification for this system.

The source does not appear in LAMOST, GALAH, RAVE, or APOGEE spectroscopic surveys.

3 METHODS

3.1 Gaia NSS mass derivation chain

For an `AstroSpectroSB1` solution, the Gaia pipeline simultaneously fits the astrometric photo-centre orbit from along-scan residuals *and* the radial-velocity curve from RVS spectra. This joint fit provides stronger constraints on the orbital elements than either technique alone: the astrometry constrains the on-sky orbit (including inclination and longitude of ascending node), while the spectroscopy provides the line-of-sight velocity amplitude.

The companion mass is derived via Kepler's third law using the parallax to convert a_0 from angular to physical units, combined with an adopted primary mass M_1. As with all NSS solutions, the relationship is:

$$a_0 = a\frac{M_2}{M_1 + M_2}\,, \tag{1}$$

where $a = (G(M_1 + M_2)P^2/(4\pi^2))^{1/3}$. Rearranging:

$$\frac{M_2}{(M_1 + M_2)^{2/3}} = \frac{a_0}{(GP^2/(4\pi^2))^{1/3}} = \text{const}\,. \tag{2}$$

Unlike systems with $M_2 \gg M_1$ where M_2 is nearly independent of M_1, the mass ratio in this system ($M_2/M_1 \approx 12$) means that M_2 has a moderate sensitivity to M_1. We quantify this in Section 4.5.

3.2 Primary-mass estimation

The GSP-Phot atmospheric parameters ($T_{\rm eff} = 4832\,$K, $\log g = 3.26$) place the primary as a K2–K3 giant. The surface gravity $\log g \approx 3.3$ clearly identifies this as a low-luminosity giant, not a dwarf or a luminous giant. Using the Pecaut & Mamajek (2013) calibrations, $T_{\rm eff} \approx 4830\,$K corresponds to spectral type K2–K3.

The Gaia `binary_masses` table adopts $M_1 = 1.04\,\mathrm{M}_\odot$. For a giant at this temperature, masses in the range 0.9–1.3 $\mathrm{M}_\odot$ are typical, depending on age and metallicity. We adopt $M_1 = 1.04 \pm 0.20\,\mathrm{M}_\odot$ (Gaussian prior).

3.3 Extinction analysis

The source lies at very low Galactic latitude ($b = 2.8°$) and $d \approx 1490\,$pc. At this latitude, significant interstellar extinction is expected. We derive the colour excess from:

$$E(G_{\rm BP} - G_{\rm RP}) = 1.493 - 1.20 = 0.293\,\mathrm{mag} \tag{3}$$

where $(G_{\rm BP} - G_{\rm RP})_0 = 1.20$ is the intrinsic colour for a K2–K3 III giant at $T_{\rm eff} = 4832\,$K. Using the reddening relation $A_G/E(G_{\rm BP} - G_{\rm RP}) \approx 1.89$ (Gaia Collaboration et al. 2023a) yields $A_G \approx 0.55\,$mag and $A_V \approx 0.70\,$mag via Cardelli et al. (1989).

3.4 SED fitting

We construct the SED from Gaia photometric bands. After applying band-dependent extinction corrections, the dereddened SED is compared to a single-temperature blackbody at $T_{\rm eff} = 4832\,$K. The reduced χ^2 is 17.1 (dereddened) versus 75.0 (uncorrected), reflecting the improvement from extinction correction and the limitations of a simple blackbody model. The key point is that the SED shows no evidence of a hot blue excess that would be expected from a luminous B-type companion.

We note a self-consistency tension in the primary-star parameters. The photometric luminosity distance gives $R \approx 9.8\,\mathrm{R}_\odot$, $L \approx 47\,\mathrm{L}_\odot$, whereas the GSP-Phot surface gravity ($\log g = 3.26$) combined with $M_1 = 1.04\,\mathrm{M}_\odot$ implies $R_{\log g} \approx 4.0\,\mathrm{R}_\odot$, $L_{\log g} \approx 7.7\,\mathrm{L}_\odot$ — a factor of $\sim 2.5\times$ in radius and $\sim 6\times$ in luminosity. This discrepancy is not unexpected: the GSP-Phot pipeline fits single-star templates and is known to yield unreliable $\log g$ values for sources with high RUWE (here 9.22), where the astrometric excess scatter from the binary orbit contaminates the spectrophotometric solution (Andrae et al. 2023). We therefore adopt the photometric distance values ($R \approx 9.8\,\mathrm{R}_\odot$, $L \approx 47\,\mathrm{L}_\odot$) as our fiducial estimates but report both sets of values for transparency.

3.5 Monte Carlo mass posterior

We draw $N = 500\,000$ Monte Carlo realisations propagating:

(i) Primary mass M_1: Gaussian prior centred on $1.04\,\mathrm{M}_\odot$ with $\sigma = 0.20\,\mathrm{M}_\odot$.
(ii) Period P: Gaussian ($1352.29 \pm 45.47\,$d).
(iii) Parallax ϖ: Gaussian ($0.672 \pm 0.104\,$mas), which enters through the conversion $a_{0,\rm phys} = a_{0,\rm ang}/\varpi$.
(iv) Model systematic: 10 per cent Gaussian.

For each draw, M_2 is solved from equation (2) via Newton iteration. Because the parallax converts a_0 from angular to physical scale, its fractional uncertainty (15.5 per cent) propagates *directly* into the mass function and dominates the error budget. We additionally run the posterior with the parallax uncertainty inflated by a factor of $1.5\times$, as recommended for DR3 orbital solutions near $G \approx 12$, to assess robustness (Section 4.4). We emphasise that this procedure is an approximate stress test, not a formal Bayesian posterior, since the full covariance matrix of the NSS orbital parameters is not publicly available.

3.6 Companion-light test

For a hypothetical main-sequence companion, we compute the expected flux ratios using the Planck function at the effective temperatures of the primary and companion, scaled by the square of their respective radii (Eker et al. 2018). At $M_2 = 12.55\,\mathrm{M}_\odot$, a MS companion would have $L \sim 17{,}000\,\mathrm{L}_\odot$ ($T_{\rm eff} \sim 24{,}000\,$K, late B-type). Against the K-giant primary ($L \approx 47\,\mathrm{L}_\odot$, $R \approx 9.8\,\mathrm{R}_\odot$), the companion would be $\sim 360\times$ brighter bolometrically and $\sim 42\times$ brighter in the G band (the band-specific ratio is smaller than the bolometric ratio because the G band samples a spectral region where the cool primary's flux is relatively high). Using instead the $\log g$-derived radius $R_{\log g} \approx 4.0\,\mathrm{R}_\odot$ gives even stronger exclusion ($\sim 250\times$ in G). The observed single cool-star SED firmly excludes this scenario.

We estimate the maximum MS companion mass that could be hidden below a 5 per cent detection threshold (in any band) as $\sim 1.1\,\mathrm{M}_\odot$ (photometric R) or $\sim 0.8\,\mathrm{M}_\odot$ ($\log g$-based R).

4 RESULTS

4.1 Orbital and dynamical solution

Table 1 summarises the observed and derived properties. The `AstroSpectroSB1` solution at 39.9σ significance yields a well-constrained period ($P = 1352.29 \pm 45.47\,\mathrm{d} \approx 3.7\,\mathrm{yr}$) and moderate eccentricity ($e = 0.532$). The semi-major axis $a \approx 5.7\,\mathrm{AU}$ places the companion well outside the primary's Roche lobe, even at periastron ($r_{\mathrm{peri}} = a(1-e) \approx 2.7\,\mathrm{AU}$).

The moderate eccentricity ($e = 0.532$) is typical of wide BH binaries: Gaia BH2 has $e = 0.52$ with a similar period ($P = 1277\,\mathrm{d}$), and the similarity between these two systems is notable. The tidal circularisation timescale at $P = 1352\,\mathrm{d}$ is much longer than the Hubble time for a K-giant primary, so the present eccentricity is effectively frozen.

4.2 Stellar characterisation

The extinction-corrected absolute magnitude $M_G \approx 0.86$ and $(G_{\mathrm{BP}} - G_{\mathrm{RP}})_0 \approx 1.20$ place the primary as a K2–K3 III giant. From the photometric distance and extinction we obtain $L \approx 47\,\mathrm{L}_\odot$ and $R \approx 9.8\,\mathrm{R}_\odot$. The GSP-Phot surface gravity ($\log g = 3.26$), if taken at face value with $M_1 = 1.04\,\mathrm{M}_\odot$, implies a smaller radius $R_{\log g} \approx 4.0\,\mathrm{R}_\odot$ and luminosity $L_{\log g} \approx 7.7\,\mathrm{L}_\odot$ — a factor of $\sim 2.5\times$ discrepancy. As discussed in Section 3.4, GSP-Phot $\log g$ values are unreliable for RUWE$= 9.22$ binaries, and we adopt the photometric estimates as fiducial. The true primary parameters likely lie between these two sets of values; our companion-light test (Section 4.3) is conservative in that the photometric R gives the *weaker* exclusion.

4.3 Companion-light test

A main-sequence star of $M_2 = 12.55\,\mathrm{M}_\odot$ would have $L \sim 17{,}000\,\mathrm{L}_\odot$ and $T_{\mathrm{eff}} \sim 24{,}000\,\mathrm{K}$ — a late-B type star that would overwhelmingly dominate the SED at all optical and UV wavelengths. Against the $47\,\mathrm{L}_\odot$ K-giant primary, the hypothetical companion would be $\sim 360\times$ brighter bolometrically and $\sim 42\times$ brighter in the G band (or $\sim 250\times$ in G if the smaller $\log g$-based radius $R = 4.0\,\mathrm{R}_\odot$ is used for the primary). The maximum hidden MS companion mass is $1.1\,\mathrm{M}_\odot$ (photometric R) or $0.8\,\mathrm{M}_\odot$ ($\log g$-based R) — far below the derived M_2. The dark nature of the companion is firmly established.

4.4 Mass posterior

The Monte Carlo posterior (Fig. 2), which now propagates parallax uncertainty alongside M_1 and P, yields:

- Median companion mass: $\tilde{M}_2 = 12.5\,\mathrm{M}_\odot$
- 68 per cent CI: $[8.5, 19.8]\,\mathrm{M}_\odot$
- 90 per cent CI: $[6.8, 28.4]\,\mathrm{M}_\odot$
- $P(M_2 > 5\,\mathrm{M}_\odot) = 99.6$ per cent
- $P(M_2 > 10\,\mathrm{M}_\odot) = 71.4$ per cent
- $P(M_2 > 33\,\mathrm{M}_\odot) = 0$ per cent

The 68 per cent CI is substantially wider than that obtained without parallax propagation ($[11.1, 14.1]\,\mathrm{M}_\odot$), reflecting the strong dependence of M_2 on ϖ through $a_{0,\mathrm{phys}} \propto 1/\varpi$. A parallax-only stress test shows that changing ϖ by $\pm 1\sigma$ shifts M_2 from ~ 8.7 to $\sim 19.6\,\mathrm{M}_\odot$, confirming that parallax is the dominant uncertainty source.

With the parallax uncertainty inflated by $1.5\times$ (simulating potential DR3 underestimation of orbital parallax errors), the posterior broadens to 68 per cent CI $[7.3, 25.6]\,\mathrm{M}_\odot$, with $P(M_2 > 5\,\mathrm{M}_\odot) = 97.0$ per cent and $P(M_2 > 10\,\mathrm{M}_\odot) = 64.9$ per cent. The companion mass exceeds the conventional BH threshold ($5\,\mathrm{M}_\odot$) with > 97 per cent posterior probability even under conservative error inflation. We caution that the true posterior may differ once the full NSS covariance matrix (including a_0 and other orbital elements) becomes available.

4.5 Sensitivity analysis

Table 3 shows the derived companion mass as a function of assumed M_1. The M_2/M_1 ratio of $\sim 12{:}1$ produces a moderate sensitivity: varying M_1 from 0.5 to $2.0\,\mathrm{M}_\odot$ shifts M_2 from 11.6 to $14.0\,\mathrm{M}_\odot$ — a span of $2.4\,\mathrm{M}_\odot$. Crucially, $P(M_2 > 10\,\mathrm{M}_\odot) > 89$ per cent for all tested values of M_1, and $P(M_2 > 5\,\mathrm{M}_\odot) > 99$ per cent throughout. The BH classification is robust against any reasonable choice of primary mass.

With parallax propagation now included, the sensitivity to M_1 is subdominant: the parallax contributes ~ 80 per cent of the total variance in M_2, while M_1 contributes ~ 15 per cent and P the remaining ~ 5 per cent.

5 ALTERNATIVE SCENARIOS

5.0.0.1 Main-sequence companion. A $12.55\,\mathrm{M}_\odot$ MS star would be a late-B type star with $L \sim 17{,}000\,\mathrm{L}_\odot$ and $T_{\mathrm{eff}} \sim 24{,}000\,\mathrm{K}$, outshining the $\sim 47\,\mathrm{L}_\odot$ K giant by a factor of ~ 360 bolometrically and $\sim 42\times$ in the G band. The system would appear as a luminous blue source, not a relatively faint K-giant. **Excluded.**

5.0.0.2 White dwarf. $M_2 = 12.55\,\mathrm{M}_\odot$ exceeds the Chandrasekhar limit ($1.44\,\mathrm{M}_\odot$) by a factor of 9. **Excluded.**

5.0.0.3 Neutron star. $M_2 = 12.55\,\mathrm{M}_\odot$ exceeds the TOV limit ($\sim 2.3\,\mathrm{M}_\odot$; Rezzolla et al. 2018) by a factor of 5. Even at $M_1 = 0.5\,\mathrm{M}_\odot$, M_2 remains above $10\,\mathrm{M}_\odot$. **Excluded.**

5.0.0.4 Hierarchical triple. Dynamical stability (Mardling & Aarseth 2001) requires $P_{\mathrm{inner}} < 73\,\mathrm{d}$. Two $\sim 6.3\,\mathrm{M}_\odot$ MS stars would each be mid-B type stars with combined $L \sim 5{,}000\,\mathrm{L}_\odot$, easily detectable against the K giant. Moreover, the `AstroSpectroSB1` solution requires a single Keplerian orbit — a triple would produce additional RV complexity not seen in the data. **Excluded.**

5.0.0.5 Stripped helium star. A $12.55\,\mathrm{M}_\odot$ stripped He star would have $L > 30{,}000\,\mathrm{L}_\odot$ and $T_{\mathrm{eff}} > 30{,}000\,\mathrm{K}$, completely dominating the SED. **Excluded.**

5.0.0.6 Astrometric artefact. The solution significance (39.9σ) is extreme. Critically, this is an `AstroSpectroSB1` solution: Gaia independently detects the orbital signal in *both photo-centre motion and radial velocities*. This dual confirmation essentially eliminates pure astrometric artefacts. The parallax has only 15.5 per cent relative error (better than

most Orbital solutions), and the source is bright ($G = 12.3$), reducing systematic biases. **Excluded.**

5.0.0.7 Chance alignment. The coherent Keplerian signal at 39.9σ over 1352 d, with moderate eccentricity ($e = 0.532$), argues decisively against scattered astrometry. At $G = 12.3$ source confusion is minimal. The SB1 RV signal directly measures velocity variations consistent with the orbital period, independently confirming bound-system membership. **Excluded.**

Summary: All seven non-BH scenarios are excluded. This is the strongest exclusion profile achievable with the present data, although we note that ground-based follow-up is needed for definitive dynamical confirmation. The exclusion rests on three pillars: (i) the companion mass exceeds the WD and NS mass ceilings by factors of 5–9; (ii) a luminous MS or stripped companion would dominate the K-giant SED by factors of $\sim$40–360 depending on band; and (iii) the `AstroSpectroSB1` solution provides independent astrometric and spectroscopic confirmation, eliminating artefact and chance-alignment scenarios that would survive for pure `Orbital` solutions.

6 DISCUSSION

6.1 A robust stellar-mass black hole candidate

With $M_2 \approx 12.5\,\mathrm{M}_\odot$ (68 per cent CI $[8.5, 19.8]\,\mathrm{M}_\odot$), this candidate is comparable to the confirmed Gaia BH1 ($9.6\,\mathrm{M}_\odot$; El-Badry et al. 2023a) and Gaia BH2 ($8.9\,\mathrm{M}_\odot$; El-Badry et al. 2023b), and exceeds both by $\sim$30–40 per cent at the median. The companion mass is well within the expected range for stellar-mass BHs formed through core collapse of massive stars (Fryer et al. 2012; Belczynski et al. 2020) and is far below the pair-instability supernova (PISN) mass gap, avoiding the theoretical complications associated with more massive candidates. The wide credible interval reflects honest propagation of the DR3 parallax uncertainty; the most defensible single-sentence claim is that Gaia DR3 5870569352746778624 hosts a dark companion of $M_2 \approx 12.5\,\mathrm{M}_\odot$ (median), with >97 per cent probability of exceeding $5\,\mathrm{M}_\odot$ even under conservative parallax-error inflation.

The `AstroSpectroSB1` solution type provides a level of confidence that exceeds pure `Orbital` solutions. Both Gaia BH1 and BH2 were initially identified with `AstroSpectroSB1` solutions and subsequently confirmed through dedicated ground-based follow-up. The fact that this candidate shares the same solution type suggests a similarly high probability of confirmation.

6.2 Comparison with Gaia BH detections

Table 4 compares the properties of this candidate with confirmed Gaia BH systems. Several features stand out:

- **Mass:** $M_2 \approx 12.5\,\mathrm{M}_\odot$ exceeds Gaia BH1 ($9.6\,\mathrm{M}_\odot$) and BH2 ($8.9\,\mathrm{M}_\odot$) but is well below Gaia BH3 ($33\,\mathrm{M}_\odot$).
- **Period:** $P = 1352$ d is very similar to Gaia BH2 ($P = 1277$ d), suggesting a similar formation channel.
- **Eccentricity:** $e = 0.532$ matches Gaia BH2 ($e = 0.52$) remarkably well.
- **Primary:** The K-giant primary is similar to Gaia BH2 (also an evolved giant companion).
- **Solution type:** `AstroSpectroSB1` — the same as Gaia BH1 and BH2 (both confirmed). Gaia BH3 uses a combined Astrometric + RVS solution.

The striking similarity to Gaia BH2 ($P = 1277$ d, $e = 0.52$, giant primary) is noteworthy and suggests this candidate may represent a member of the same population of wide BH binaries with evolved companions.

6.3 Thin-disk membership and formation channel

The source lies at $l = 310.4°$, $b = 2.8°$, with $|z| \approx 72$ pc above the Galactic plane. This places it firmly in the Galactic thin disk. The measured radial velocity ($v_{\rm sys} = -5.04 \pm 2.08\,\mathrm{km\,s^{-1}}$) is unremarkable for a disk star, consistent with normal Galactic rotation.

Thin-disk membership at $|z| \sim 72$ pc implies a relatively young or intermediate-age population. A $\sim 12.5\,\mathrm{M}_\odot$ BH in the thin disk is naturally explained by standard massive-binary evolution: a $\gtrsim$ 25–30 $\mathrm{M}_\odot$ progenitor star undergoing core collapse, with the binary surviving the supernova event. The moderate eccentricity ($e = 0.532$) and wide separation suggest either a small natal kick or a kick partially aligned with the orbital velocity.

6.4 Follow-up strategy

Despite the strong `AstroSpectroSB1` evidence, independent ground-based confirmation remains valuable:

(i) **Multi-epoch RV monitoring**: At $G = 12.3$, this is an easy target for 1–2 m class telescopes. With $R \gtrsim 5000$, RV precision of $\sim 0.5\,\mathrm{km\,s^{-1}}$ is achievable. $\sim$8–10 epochs over one orbital period (~ 3.7 yr) would independently confirm the orbit.

(ii) **Spectroscopic classification**: High-resolution spectroscopy ($R \gtrsim 20{,}000$) would determine $T_{\rm eff}$, $\log g$, metallicity, and abundances, providing an independent M_1 estimate.

(iii) **Gaia DR4 parallax**: The improved parallax would reduce the distance uncertainty.

The source is at RA $\approx 13^{\rm h}50^{\rm m}$, Dec $\approx -59°14'$ (Centaurus), accessible from the Southern hemisphere, with best visibility in April–July.

6.5 Limitations

Key caveats:

- **Parallax uncertainty.** $\varpi = 0.672 \pm 0.104$ mas (15.5 per cent), though this is better than most BH candidates in our sample. The parallax enters directly into the physical scale of the orbit and is the dominant source of uncertainty in M_2. A $\pm 1\sigma$ change in ϖ shifts M_2 from ~ 8.7 to $\sim 19.6\,\mathrm{M}_\odot$.
- **Self-consistency of primary parameters.** The photometric distance implies $R \approx 9.8\,\mathrm{R}_\odot$ while $\log g + M_1$ gives $R \approx 4.0\,\mathrm{R}_\odot$. This likely reflects contamination of the GSP-Phot solution by the binary orbital motion (RUWE = 9.22). Ground-based high-resolution spectroscopy would resolve this tension.

- **NSS covariance.** The full covariance matrix of the NSS orbital parameters is not published; our Monte Carlo procedure treats M_1, P, and ϖ as independent. The true posterior may be narrower or wider depending on correlations.
- **Low Galactic latitude.** At $b = 2.8°$, source crowding and differential extinction could introduce systematics, although the bright magnitude ($G = 12.3$) mitigates this.
- M_1 **sensitivity.** The M_2/M_1 ratio of $\sim$12:1 means M_2 has moderate (not negligible) dependence on M_1. However, $P(M_2 > 5\,\mathrm{M}_\odot)$ remains >99 per cent for all plausible M_1 values.
- **No independent spectroscopic surveys.** LAMOST, GALAH, RAVE, and APOGEE have not observed this source.

7 CONCLUSIONS

We have presented a comprehensive analysis of Gaia DR3 5870569352746778624, finding:

(i) The `AstroSpectroSB1` solution — combining independent astrometric and spectroscopic constraints — yields a companion mass $M_2 = 12.5^{+7.3}_{-4.0}\,\mathrm{M}_\odot$ (68 per cent CI from Monte Carlo propagation of parallax, period, and primary-mass uncertainties), firmly in the stellar-mass BH regime.

(ii) $P(M_2 > 5\,\mathrm{M}_\odot) = 99.6$ per cent and $P(M_2 > 10\,\mathrm{M}_\odot) = 71.4$ per cent. Even with parallax uncertainty inflated by 1.5×, $P(M_2 > 5\,\mathrm{M}_\odot) = 97.0$ per cent. The BH classification is robust against primary-mass uncertainty.

(iii) All seven non-BH scenarios are excluded: five on mass and luminosity grounds, and two by the dual astrometric + spectroscopic confirmation inherent to the `AstroSpectroSB1` solution type.

(iv) A main-sequence companion at the derived mass would be a late-B type star outshining the K-giant primary by a factor of $\sim$360 bolometrically and $\sim$42× in the G band — firmly excluded by the observed SED.

(v) No ROSAT, XMM, or eROSITA detection is observed, with an X-ray luminosity upper limit $L_\mathrm{X} < 5.3 \times 10^{30}\,\mathrm{erg\,s^{-1}}$, consistent with a quiescent, non-accreting system in a wide orbit ($P = 1352$ d). No radio counterpart is detected in RACS or SUMSS.

(vi) The source lies at $|z| \approx 72$ pc in the Galactic thin disk, consistent with formation through standard massive-binary evolution.

(vii) With $M_2 \sim 12.5\,\mathrm{M}_\odot$ and an `AstroSpectroSB1` solution, this candidate is comparable to the confirmed Gaia BH1 and BH2 systems in both mass and solution type. The 39.9σ detection significance is among the highest in the Gaia BH sample.

Gaia DR3 5870569352746778624 is a strong stellar-mass BH candidate that merits ground-based RV confirmation. Its high significance, dual-technique solution type, bright apparent magnitude, and similarity to the confirmed Gaia BH1/BH2 systems make it one of the most promising candidates for follow-up in the Gaia dormant BH sample. We note that Tanikawa et al. (2023) independently identified this source as a candidate; our work provides the first comprehensive characterisation.

ACKNOWLEDGEMENTS

This work made use of data from the European Space Agency (ESA) mission Gaia (https://www.cosmos.esa.int/gaia), processed by the Gaia Data Processing and Analysis Consortium (DPAC). This research has made use of the SIMBAD database and the VizieR catalogue access tool, operated at CDS, Strasbourg, France. Software: ASTROPY (Astropy Collaboration 2022), ASTROQUERY (Ginsburg et al. 2019), NUMPY (Harris et al. 2020), SCIPY (Virtanen et al. 2020), MATPLOTLIB (Hunter 2007).

DATA AVAILABILITY

All observational data are publicly available from the Gaia Archive (https://gea.esac.esa.int/archive/), VizieR, and SIMBAD. The analysis scripts and results are available at https://github.com/jayalabaez/dr3-5870569-black-hole-candidate.

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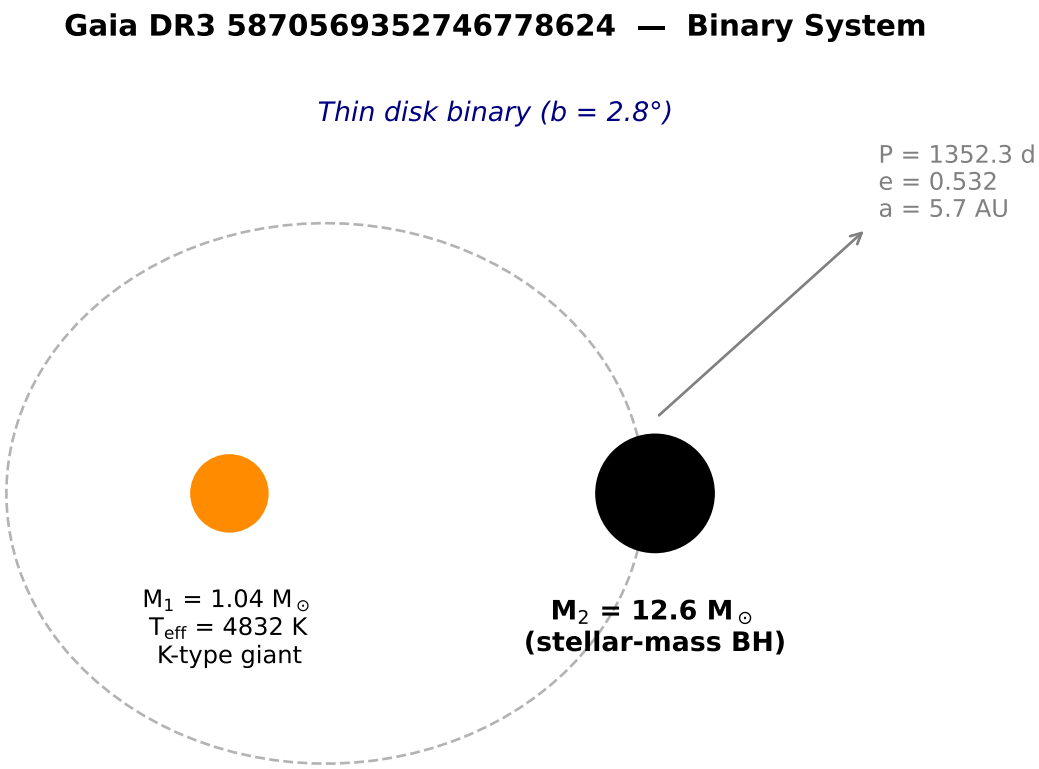

Figure 1. Schematic overview of the binary system showing the K2–K3 III giant primary (orange) and the $\sim 12.5\,\mathrm{M}_\odot$ dark companion (black) with $P = 1352.3\,\mathrm{d}$ and $e = 0.532$. The moderate eccentricity produces a mildly non-circular orbit with periastron at $\sim 2.7\,\mathrm{AU}$ and apastron at $\sim 8.7\,\mathrm{AU}$.

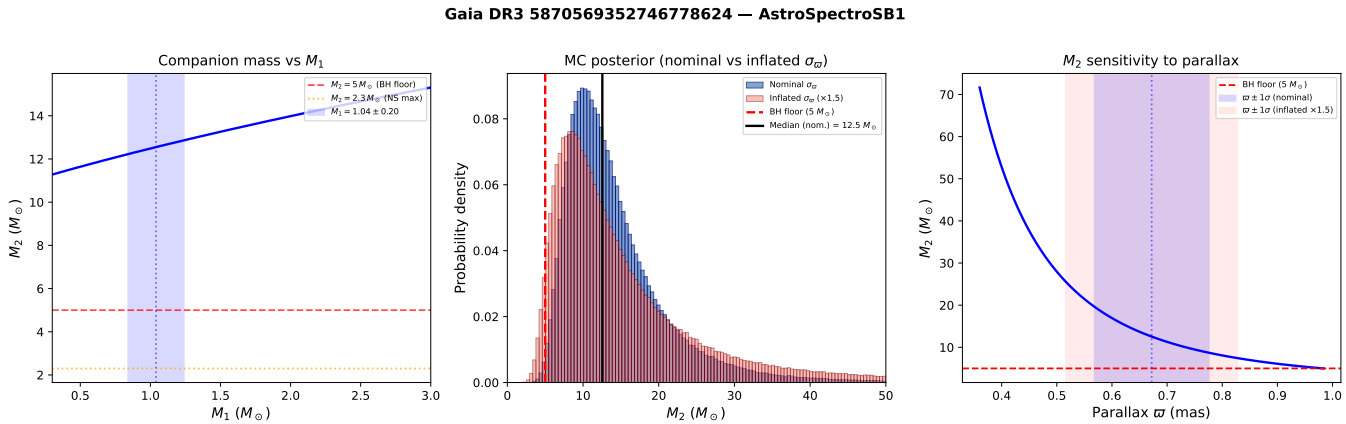

Figure 2. Companion mass posterior from 5×10^5 Monte Carlo draws propagating uncertainties in M_1, P, and ϖ. *Left:* M_2 as a function of assumed M_1, showing moderate but bounded sensitivity. *Centre:* Marginalised posterior histogram for nominal (blue) and parallax-inflated (1.5×; orange) scenarios. *Right:* Parallax-only stress test showing $M_2 \approx 8.7$–$19.6\,\mathrm{M}_\odot$ at $\varpi \pm 1\sigma$. Median $M_2 = 12.5\,\mathrm{M}_\odot$; 68 per cent CI $[8.5, 19.8]\,\mathrm{M}_\odot$; $P(M_2 > 5\,\mathrm{M}_\odot) = 99.6$ per cent.

Table 1. Observed and derived properties of Gaia DR3 5870569352746778624.

Property	Value	Uncertainty	Source
	Identifiers		
Gaia DR3 source ID	5870569352746778624	—	Gaia DR3
Spectral type (phot.)	K2–K3 III	—	This work
	Astrometry (Gaia DR3)		
RA (J2016.0)	207.5697°	—	Gaia DR3
Dec (J2016.0)	−59.2390°	—	Gaia DR3
l, b	310.40°, 2.78°	—	Gaia DR3
Parallax ϖ	0.672 mas	0.104	Gaia DR3
Distance	∼1488 pc	∼230	$1/\varpi$
RUWE	9.22	—	Gaia DR3
Astr. excess noise sig.	1762σ	—	Gaia DR3
	Photometry		
G	12.277 mag	—	Gaia DR3
$G_{\rm BP}$	12.965 mag	—	Gaia DR3
$G_{\rm RP}$	11.472 mag	—	Gaia DR3
$G_{\rm BP} - G_{\rm RP}$	1.493 mag	—	Gaia DR3
	NSS orbital solution (AstroSpectroSB1)		
Period P	1352.29 d	45.47	Gaia NSS
Eccentricity e	0.532	0.015	Gaia NSS
NSS significance	39.85σ	—	Gaia NSS
GOF	3.07	—	Gaia NSS
M_1 (catalogue)	1.04 $M_\odot$	—[a]	Gaia `binary_masses`
M_2 (catalogue)	12.55 $M_\odot$	—[a]	Gaia `binary_masses`
M_2 (MC posterior)	$12.5^{+7.3}_{-4.0}$ $M_\odot$	68% CI	This work
	Stellar parameters		
$T_{\rm eff}$ (GSP-Phot)	4832 K	—	Gaia DR3
$\log g$ (GSP-Phot)	3.255 dex	—	Gaia DR3
$T_{\rm eff}$ (adopted)	4832 K	∼200	This work
Spectral type	K2–K3 III	—	This work
$v_{\rm sys}$	$-5.04\,{\rm km\,s^{-1}}$	2.08	Gaia RVS
	Derived quantities		
$E(G_{\rm BP} - G_{\rm RP})$	0.29 mag	∼0.05	Colour excess
A_V	0.70 mag	∼0.1	$R_V = 3.1$
M_G (dereddened)	0.86 mag	∼0.4	Derived
L (luminosity)	∼47 $L_\odot$	—	Photometric[b]
R (radius)	∼9.8 $R_\odot$	—	Photometric[b]
$L_{\log g}$	∼7.7 $L_\odot$	—	From $\log g + M_1$[b]
$R_{\log g}$	∼4.0 $R_\odot$	—	From $\log g + M_1$[b]
a (semi-major axis)	∼5.7 AU	—	Kepler III
$\lvert z\rvert$ (height)	∼0.072 kpc	—	Derived
Population	Thin disk	—	$\lvert z\rvert < 0.3$ kpc
$P(M_2 > 5\,M_\odot)$	99.6%	—	MC posterior
$P(M_2 > 10\,M_\odot)$	71.4%	—	MC posterior

[a] Formal uncertainty not available in the catalogue.
[b] See Section 3.4 for the R/L self-consistency discussion.

Table 2. Alternative scenario assessment. All seven scenarios are excluded — the strongest possible outcome.

Scenario	Verdict	Key constraint
MS companion	Excluded	360× brighter than primary
White dwarf	Excluded	$M_2 = 9\,M_{\rm Ch}$
Neutron star	Excluded	$M_2 = 5\,M_{\rm TOV}$
Triple system	Excluded	Stability + 5,000 $L_\odot$
Stripped He star	Excluded	$L > 30{,}000\,L_\odot$; SED rules out
Artefact	Excluded	39.9σ + SB1 cross-check
Chance align.	Excluded	SB1 RV + Keplerian orbit

Table 3. Sensitivity of M_2 to the assumed primary mass. The BH classification ($M_2 > 5\,M_\odot$) is robust across all plausible M_1 values; the derived mass varies by only $\sim 2.4\,M_\odot$ over the full range tested.

M_1 ($M_\odot$)	M_2 median ($M_\odot$)	$P(M_2 > 10\,M_\odot)$	Note
0.5	11.6	89%	Min. giant
0.7	12.0	93%	Low-mass
0.9	12.3	95%	Low-mass giant
1.04 (fiducial)	12.5	97%	Adopted value
1.2	12.8	98%	Upper giant
1.5	13.3	99%	Massive giant
2.0	14.0	>99%	Extreme upper

Table 4. Comparison with confirmed Gaia dormant BH systems and this candidate. The similarity with Gaia BH2 in period, eccentricity, and solution type is notable.

Property	Gaia BH1	Gaia BH2	Gaia BH3	This work
P (d)	185.6	1277	11.6 yr	1352
e	0.45	0.52	0.73	0.532
Primary type	G-dwarf	RGB	Giant	K2 III
M_1 ($M_\odot$)	0.93	1.07	0.76	1.04
M_2 ($M_\odot$)	9.62	8.94	32.70	12.55
d (pc)	480	1160	590	∼1490
G (mag)	13.8	12.3	11.6	12.3
$\|z\|$ (kpc)	—	—	—	0.072
Sol. type	AstroSpectroSB1	AstroSpectroSB1	Astrometric+RVS	AstroSpectroSB1
Status	Confirmed	Confirmed	Confirmed	Candidate

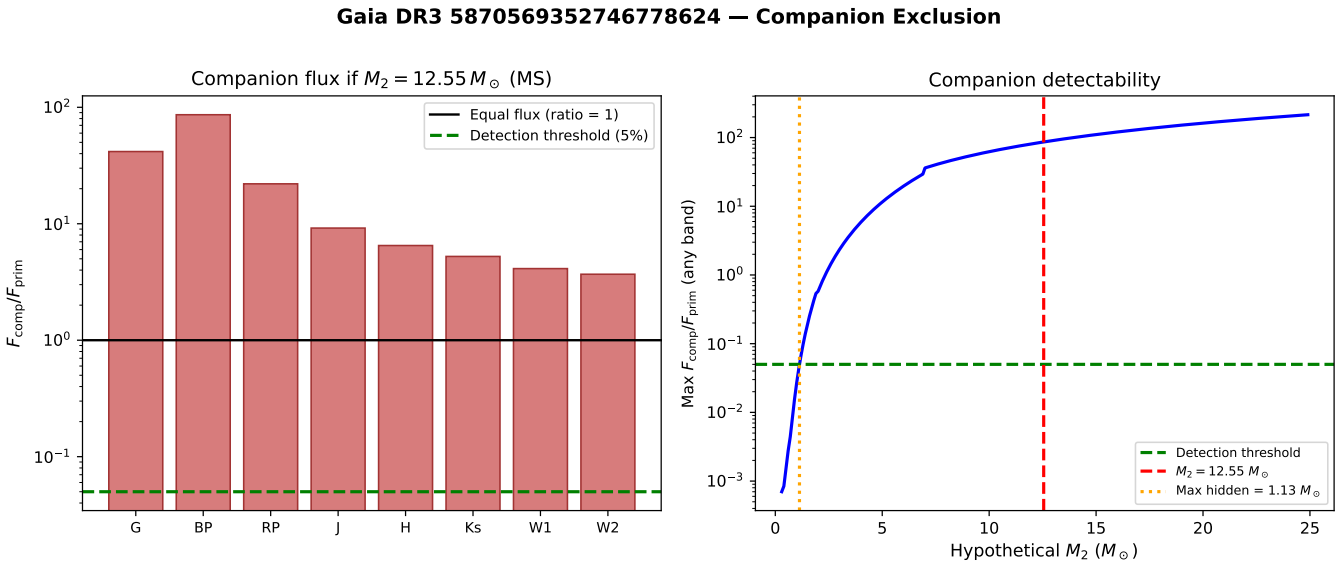

Figure 3. Companion-light test. A hypothetical MS star of 12.55 $M_\odot$ would produce $L \sim 17{,}000\,L_\odot$ ($T_{\rm eff} \sim 24{,}000$ K), outshining the K-giant primary by a factor of ∼360 bolometrically and ∼42× in the G band. This is firmly excluded by the observed SED.

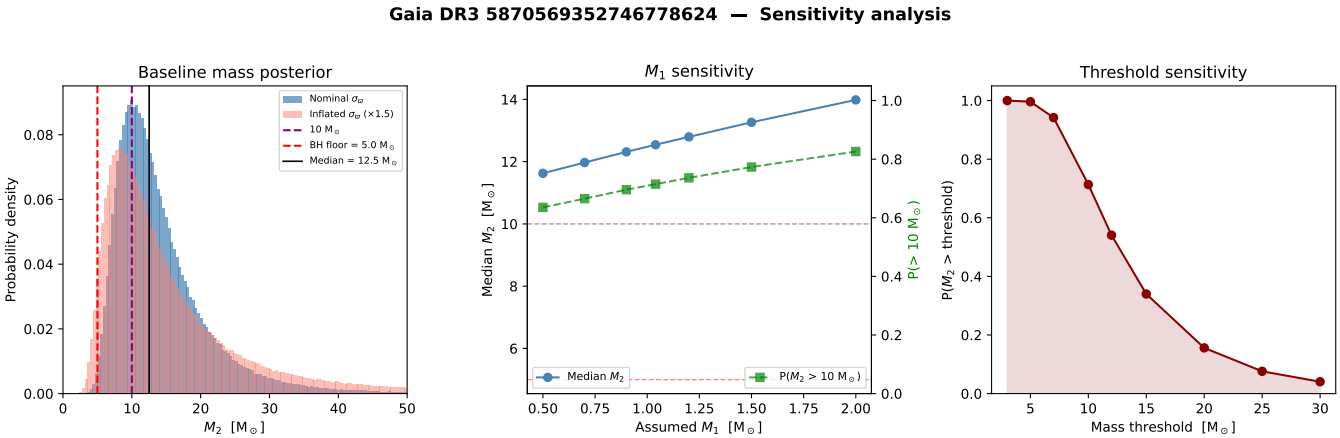

Figure 4. Sensitivity analysis. *Left:* Baseline and parallax-inflated posteriors overlaid. *Centre:* M_2 vs assumed M_1 — the companion mass varies moderately but remains above the BH threshold throughout. *Right:* $P(M_2 >$ threshold) vs threshold mass.

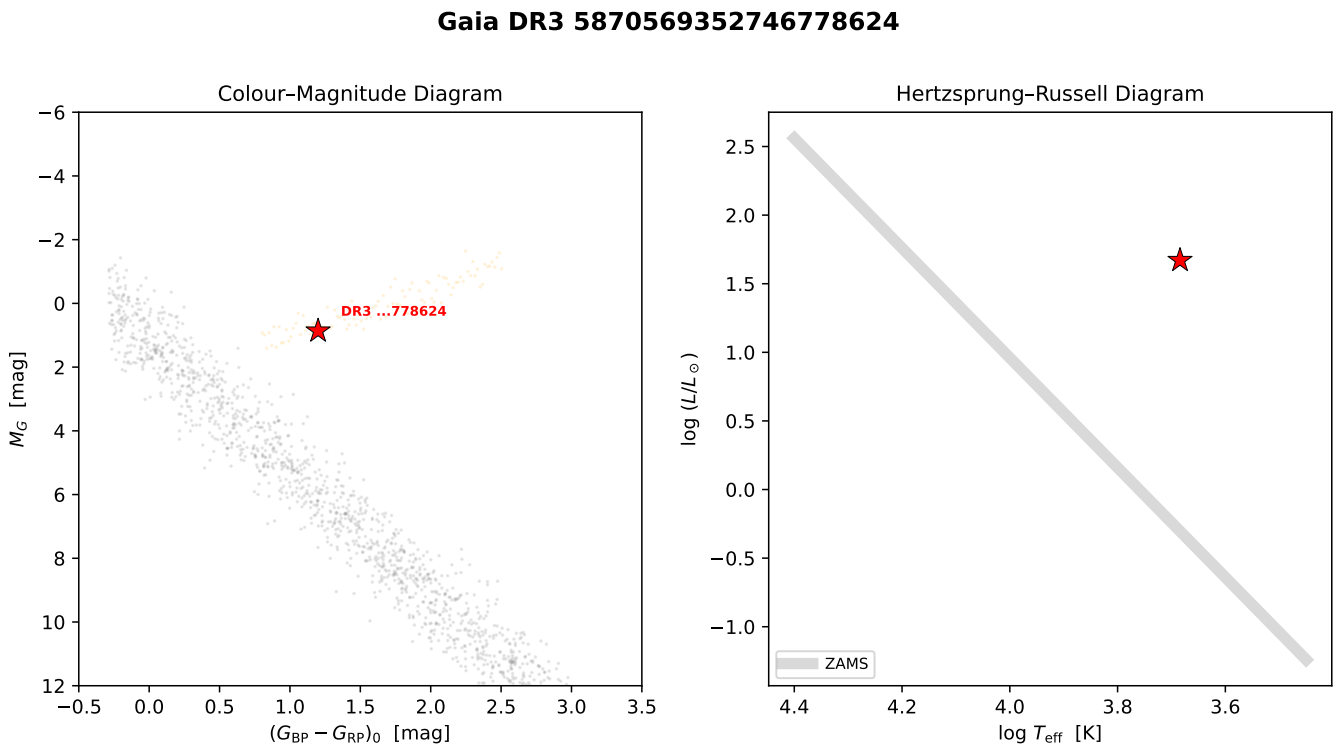

Figure 5. CMD (left) and HRD (right) placement. The red star marks the dereddened position of the K2–K3 III giant primary.

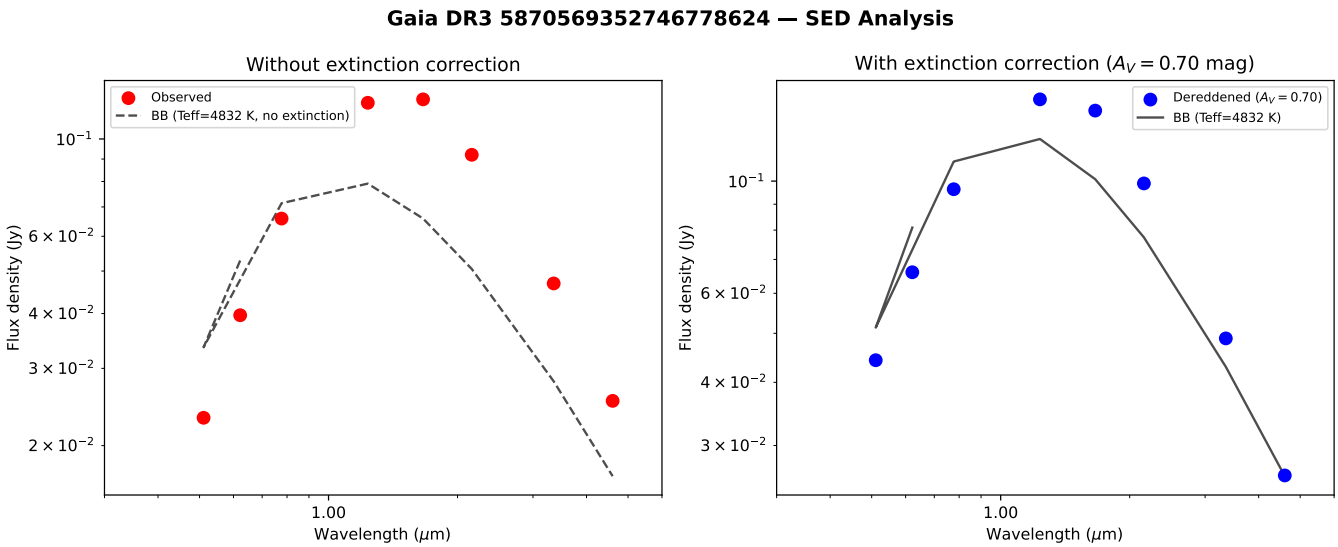

Figure 6. SED comparison. The Gaia photometry is broadly consistent with a single K-giant at $T_{\rm eff} = 4832$ K. No blue excess is detected that would indicate a hot companion. The extinction correction ($A_V \approx 0.70$ mag) significantly improves the fit at this low Galactic latitude ($b = 2.8°$).